

COMPSCI 1JC3
Introduction to Computational Thinking
Fall 2026

Course Outline

Dr. William M. Farmer
McMaster University

Revised: September 6, 2026

Note: This course outline contains important information that may affect your grade. You should retain it and refer to it throughout the term, as you will be assumed to be familiar with the rules specified in this document.

Instructor

Dr. William M. Farmer

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Office Hours: To see me in person or via a Microsoft Teams call, please send me an email message with some times you are free.

Teaching Assistants (TAs)

Jason Balaci	balacij@mcmaster.ca	Graduate TA
Brandon Bosman	bosmanb@mcmaster.ca	Graduate TA
Neaha Bijo	bijon@mcmaster.ca	Undergraduate TA
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Hetvi Parikh	parikh5@mcmaster.ca	Undergraduate TA
Christina Sauro	sauroc@mcmaster.ca	Undergraduate TA
Tony Yuan	yuant15@mcmaster.ca	Undergraduate TA

Teaching Schedule

Lecture C01/C03	TuThFr	08:30–09:20	Dr. Farmer
Lecture C02/C04	TuWeFr	12:30–13:20	Dr. Farmer
Tutorial T01	Mo	16:30–18:20	Hetvi
Tutorial T02	Mo	12:30–14:20	Brandon
Tutorial T03	Tu	10:30–12:20	Tony
Tutorial T04	Tu	10:30–12:20	Hassan
Tutorial T05	We	16:30–18:20	Brandon
Tutorial T07	Th	16:30–18:20	Nora
Tutorial T08	Th	16:30–18:20	Neaha
Tutorial T09	Mo	16:30–18:20	Christina

Course Web Site

This course will be administered via *Avenue to Learn* (*Avenue* for short). Go to

<http://avenue.mcmaster.ca/>.

to access the course’s Avenue page. Please send only normal email to the instructional staff; do not send mail via Avenue.

Students should be aware that, when they access the electronic components of this course, private information such as first and last names, user names for the McMaster email accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in this course will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with your Instructor.

It is the student’s responsibility to be aware of the information on the course’s Avenue pages and to check regularly for announcements.

Course Schedule

The course schedule is given in a separate document entitled *Course Schedule*.

Calendar Description

“Exploration of thinking that is inspired, supported, and enabled by computing. Survey of the salient ideas, methods, and technologies in the major areas of computing including basic data types, logic, operating systems, computer networking, web computing, information security, digital media, software development, and problem solving techniques. Introduction to the fundamentals of functional programming.”

Mission

The *mission* of the course is to (1) introduce students to *computational thinking* (i.e., thinking that is inspired, supported, or enabled by computing) and (2) help students begin the process of developing a sophisticated understanding of *computing*. The students will achieve these goals by exploring the salient ideas, methods, and technologies in computing and by solving problems using functional programming.

Learning Objectives: Precondition

A *learning objective* for a course is something the student is expected to know and understand or to be able to do. The *precondition* of the course is the set of university-level learning objectives that the student is expected to have achieved before the start of the course. Since this course has no university-level prerequisites, its precondition is empty.

Learning Objectives: Postcondition

The *postcondition* of the course is the set of learning objectives that the student is expected to have achieved by the end of the course. The postcondition for this course is given below.

1. Students should know and understand:
 - a. What computational thinking is.
 - b. The key ideas, methods, technologies, and areas of computing.
 - c. Functional programming language concepts.
2. Students should be able to:
 - a. Design and write a simple functional program that satisfies a simple requirements specification.
 - b. Formulate a test plan for a simple functional program.
 - c. Analyze a simple functional program.
 - d. Use Haskell as a problem-solving tool.

By the end of the course, each student should be comfortable programming in Haskell.

Textbooks

1. Paul S. Wang, **From Computing to Computational Thinking** [abbreviated **CCT**], Chapman and Hall/CRC, 2015. ISBN-13: 978-1482217650. Available in paperback on Amazon.ca for \$108.44 CAD. See the reading assignments in the *Course Schedule*. **Optional but strongly recommended.**

2. Simon Thompson, **Haskell: The Craft of Functional Programming (3rd Edition)** [abbreviated **HCFP**], Addison-Wesley, 2011. ISBN-13: 978-0201882957. Available in paperback on Amazon.ca for \$19.98 CAD. See the reading assignments in the *Course Schedule*. **Optional but strongly recommended.**
3. Miran Lipovača, **Learn You a Haskell for Great Good!** [abbreviated **LHGG**], No Starch Press, 2011, ISBN-13: 978-1593272838. Available in paperback on Amazon.ca for \$64.67 CAD. **Optional.**

Note: Amazon.ca prices were recorded on August 24, 2026.

Other Resources

1. **iClicker Student App** enables several kinds of in-class participation including attendance recording. A 6-month subscription costs \$15.99 CAD for six months and \$24.99 CAD for one year. **Required.**
2. **GHCup** is the tool for installing the **Glasgow Haskell Compiler (GHC)** and **Stack** Haskell management tool. Available for free. **Required.**
3. **Visual Studio Code (VS Code)** with the **Haskell language support plugin** is the recommended code editor and IDE (integrated development environment) for programming in Haskell. Available for free. **Optional but recommended.**
4. **Microsoft Teams** may be used to conduct some discussion sessions, tutorials, and office hours. Login with your MacID. Available for free.

Work Plan

There will be lectures, discussion sessions, tutorials, a discussion forum, M&Ms, a bio sheet, programming exercises, programming assignments, two midterm tests, and a final exam. The material in the entire text of CCT and in Chapters 1–13 of HCFP will be covered. Each student is expected to devote a minimum of nine hours of work per week for this course.

The lectures for the course will be video presentations with slides given by the Instructor. The slides and a link to the video presentations are posted on Avenue. See the lecture video assignments in the *Course Schedule*.

Three 50-minute, in-person discussion sessions (which are scheduled during the lectures times each week on Tuesdays, Thursdays, and Fridays for the C01/C03 section and on Tuesdays, Wednesdays, and Fridays each week for the C02/C04 section) will be led by the Instructor. During the discussion sessions, students will be selected from a randomized class list and asked questions about the material covered in the lectures and reading assignments for the week. If a student is asked a question during a discussion session, the student will receive a mark of 0 if the student is not present and a mark between 0.75 and 1.25 if the student is present. The student will

receive 0.75 for no answer or a poor answer; 1.0 for a good answer (which is considered a full mark); and 1.25 for an excellent answer (full mark plus 25% bonus). During each round of questions, if the student is not present when asked a question, the student will automatically be given a second chance to answer a question on a later date.

One 110-minute tutorial on programming in Haskell will be led by a TA. There will be video presentations with slides that the students should watch before the tutorials. The slides and a link to the video presentations are posted on Avenue. See the tutorial video assignments in the *Course Schedule*.

The students are required to participate in a discussion forum located on Avenue under Communication / Discussions. Students are encouraged to use the forum to discuss the course material and to ask and answer questions. The Instructor and TAs will monitor the forum and participate in the discussion as needed.

At the end of each week, the students are required to submit a short paragraph (2–4 sentences) that (1) describes something from the week’s lectures, discussion sessions, reading, or tutorial that was especially meaningful or memorable to the student, (2) identifies something that was not understood by the student, or (3) comments on how the student thinks the course is going. These *meaningfuls and memorables (M&Ms)* are intended to help the student to reflect on what they are learning and to give the Instructor feedback on what the students are experiencing in the course. The M&Ms are submitted to a discussion on Avenue under Communication / Discussions.

Each student is required to submit a bio sheet about themselves using their Instructor’s bio sheet as a model (which is posted on Avenue under Content / Course Information).¹ Bio sheets are submitted as an assignment on Avenue under Assessments / Assignments / Bio Sheets.

There will be eight sets of programming exercises that will be discussed in the tutorials. Students will be required to submit their solutions for a set of exercises during the weeks when an assignment is not due. The exercises will be marked automatically. See the exercises due dates in the *Course Schedule*.

There will be four programming assignments that will be marked by the TAs. The mark for an assignment will be determined in part by running the code and in part by reading the code. Each programming assignment will be due approximately two weeks after it is posted on Avenue. See the assignment due dates in the *Course Schedule*.

There will be two 120-minute midterm tests, one on Thursday, October 22, 2026 at 19:00–21:00 PM and another on Thursday, November 26, 2026 at 19:00–21:00 PM. The final exam, which will cover the material of the entire course, will be 2.5 hours long. It will take place on the date scheduled by the University. The midterms and final exam are multiple choice and *closed book* (which means you may not refer to any notes or books during

¹I would like to thank Dr. Lydell Wiebe for communicating the bio sheet idea to me.

the exam period).

The midterm tests will be done in a two-stage manner. Each student will take the test twice. For the first stage the student will do the test by themselves without help from anyone. For the second stage the student will do the test again with help from any of the students in the testing room. 90 minutes will be allotted for the first stage and the remaining 30 minutes for the second stage. The first stage test counts for 80% of the test mark and the second stage counts for 20%.

Course Topics

- 01 What is Computational Thinking?
- 02 What is Functional Programming?
- 03 Numbers.
- 04 Logic.
- 05 Recursion.
- 06 Algebraic Data Types.
- 07 Operating Systems and the Internet.
- 08 The World Wide Web.
- 09 Information Security.
- 10 Problem Solving.
- 11 Data.
- 12 Software Development.

Midterm Course Review

In the middle of the term, the students will have the opportunity to complete an anonymous survey on Avenue about how the course is going. In addition, each student will be invited to attend a course review session in which the student can ask their Instructor questions about the course and give feedback directly to the Instructor. The course review sessions will be held soon in Week 07 (October 26–29). Attendance is optional, but each student who participates in one of the course review sessions will receive a 1.0 percentage point bonus. The feedback that is received from the course survey and course review sessions may be used to modify how the course is working.

End-of-Term Student Course Experience Survey

Near the end of the term, each student will have the opportunity to evaluate online the effectiveness of this course. The feedback that is received from the course evaluation is very valuable to the Instructor and will be used to improve the course in subsequent years.

Each student will receive a bonus based on the level of class participation in the end-of-term Student Course Experience Survey according to the following table:

Class Participation	Bonus
60–64%	0.25
65–69%	0.50
70–74%	0.75
75–79%	1.00
80–84%	1.25
85–89%	1.50
90–94%	1.75
95–100%	2.00

Thus, for example, if 75% of the students enrolled in CS 1JC3 participate in the course evaluation of CS 1JC3, every student's final mark will receive a 1.0 percentage point bonus.

Drop-In Centre in ITB 242

To help you with any questions you may have on programming in Haskell and other languages, a computing Drop-In Centre is offered by the Department of Computing and Software in ITB 242 Monday to Friday at 9:30–16:20. The Drop-In Centre is run by John Nakamura (nakamura@mcmaster.ca). If you need any help with Haskell programming, please take advantage of this valuable resource.

Discrimination

The Faculty of Engineering is concerned with ensuring an environment that is free of all adverse discrimination. If there is a problem, that cannot be resolved by discussion among the persons concerned, individuals are reminded that they should contact their Department Chair and the *Human Rights and Equity Services (HRES)* office as soon as possible.

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g., the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university.

For information on the various types of academic dishonesty please refer to the *Academic Integrity Policy*, located at

<https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/>.

The following illustrates only three forms of academic dishonesty:

1. Plagiarism, e.g., the submission of work that is not one’s own or for which other credit has been obtained.
2. Improper collaboration in group work.
3. Copying or using unauthorized aids in tests and examinations.

Course-specific regulations:

1. *Your work must be your own.* Plagiarism and copying will not be tolerated! If it is discovered that you plagiarized or copied, it will be considered as academic dishonesty. In particular, submitting work produced by ChatGPT, Claude, Gemini, Copilot, and other generative AI systems will be considered plagiarism.
2. *You may not share your code.* You may not give other students a printed or electronic copy of any part of your code. You may not upload your code to a public web site or repository.
3. *Students may be asked to defend their written work orally.*

Authenticity/Plagiarism Detection

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g., A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., online search, other software, etc.). For more details about McMaster’s use of Turnitin.com please go to

<http://www.mcmaster.ca/academicintegrity/>.

In this course, the MOSS (Measure Of Software Similarity) system, described at

<https://theory.stanford.edu/~aiken/moss/>,

may be used to check for plagiarism in the assignments submitted by students.

Courses with an Online Element

Some courses may use online elements (e.g., email, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster email accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses online elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning, and working communities. These expectations are described in the *Code of Student Rights & Responsibilities* (*Code* for short), located at

<https://secretariat.mcmaster.ca/app/uploads/Code-of-Student-Rights-and-Responsibilities.pdf>.

All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable

participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

In this course, students are expected to be respectful and professional when they are participating in the discussion sessions, tutorials, discussion forum, and M&Ms. Moreover, students are welcome to use chat during meetings via Microsoft Teams, but they must restrict their discussion to topics that are relevant to the course. Any student that violates these rules in the discussion sessions shall be permanently removed from the list of participants for the discussion sessions. Similarly for the tutorials, discussion forum, and M&Ms.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact *Student Accessibility Services (SAS)* at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's *Academic Accommodation of Students with Disabilities* policy.

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO policy](#). Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, exercises, assignments, and tests.

Requests for Relief for Missed Academic Term Work

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the the [McMaster Student Absence Form Policy \(MSAF Policy\)](#).

A student who would like to receive accommodation for missed academic work due to an absence needs to complete a McMaster Student Absence Form (MSAF) online at

<https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/msaf-policy-2/resource-page/>.

When the MSAF tool asks you for the party who should receive your request for accommodation, enter your instructor's email address,

wmfarmer@mcmaster.ca. MSAFs sent to any other email address will be ignored.

Course-specific regulations:

1. The accommodation for a missed exercises or assignment will be that the due date is moved forward in time exactly five days.
2. The accommodation for a single missed midterm test will be that the midterm test is not required and the weight of the midterm is added to the weight of the final exam.
3. MSAFs are not accepted for missed class participation marks (discussion session questions, discussion session attendance, tutorial attendance, Discussion Forum, M&Ms, and bio sheets).

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical, and artistic work, including lectures by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.

AI Policy

Students are allowed to use AI in this course, but it is recommended that AI is used as little as possible. *However, as mentioned above, submitting work produced by ChatGPT, Claude, Gemini, Copilot, and other generative AI systems will be considered plagiarism. It is also highly recommended that students do not use AI in any manner to help them learn how to program in Haskell or to do the exercises and assignments.*

Other Policy Statements

1. Significant study and reading outside of class is required.
2. The student is expected to participate in the discussions during discussions sessions and tutorials.
3. Exercises, assignments, and M&Ms may not be submitted late and the midterm tests may not be taken later without *prior* approval from the Instructor.
4. If there is a problem with the marking of a programming assignment, the student should first discuss the problem with the TA who marked it. Assignment marks will only be changed if the problem is reported within two weeks of the date that the assignment mark was returned.
5. The use of notes and books is *not* permitted during the midterm tests and final exam.
6. No electronic devices (calculators, cell phones, smart eye glasses, etc.) may be used during midterm tests and the final exam. They will be confiscated if they are within reach of a student, whether or not they have been actually used.
7. Email with a source address outside of McMaster University will not be read by the instructional staff.
8. Suggestions on how to improve the course and the Instructor's teaching methods are always welcomed.

Marking Scheme

The course grade will be based on the student's performance on class participation, programming exercises, programming assignments, midterm tests, and final exam as follows:

Class participation:

1. Discussion session questions	6%
2. Discussion session attendance	2%
3. Tutorial attendance	2%
4. Discussion forum	2%
5. Meaningfuls and memorables (M&Ms)	2%
6. Bio sheet	2%

Programming:

1. Programming exercises (8)	8%
2. Programming assignments (4)	16%

Exams:

1. Midterm test 1	16%
2. Midterm test 2	16%
3. Final exam	28%

Total: **100%**

Bonuses:

1. Maximum discussion session questions bonus	1.5%
2. Midterm course review session bonus	1%
3. Maximum course experience survey bonus	2%
4. McMaster Engineering Competition (MEC)	1%

Notes:

1. The Instructor reserves the right to adjust the marks for a programming assignment, midterm test, or final exam by increasing or decreasing every score by a fixed number of points.
2. The *discussion session questions mark* is x with $0 \leq x \leq 6$ where x is the average mark a student receives for questions that the Instructor asks the student in discussion sessions multiplied by 6. A mark of 1.0 is a full mark, while a mark of 1.25 is a full mark plus a 25% bonus. During each round of questions, if the student is not present when asked a question, the student will automatically be given a second chance to answer a question on a later date.

3. The *discussion session attendance mark* is x with $0 \leq x \leq 2$ where x is the percentage of discussion sessions the student attended multiplied by 2. So if you attend all discussion sessions your mark is 2.0.
4. The *tutorial attendance mark* is x with $0 \leq x \leq 2$ where x is the percentage of tutorials the student attended multiplied by 2. So if you attend all tutorials your mark is 2.0.
5. The *discussion forum mark* is x with $0 \leq x \leq 2$ where x is the number of postings (questions or answers) made on different days to the discussion forum divided by 10. So if you have 15 postings your mark is 1.5, and if you have 20 or more postings your mark is 2.0. Postings must be professional and germane to the subjects studied to receive credit.
6. The *M&Ms mark* is x with $0 \leq x \leq 2$ where x is the percentage of acceptable weekly M&Ms that the student submitted multiplied by 2. So if you submitted 10 of the 13 weekly M&Ms your mark is 1.54. Submitted M&Ms must be professional and germane to the subjects studied to receive credit.
7. The *bio sheet mark* is 0 if it is not submitted; 1 if submitted after October 11; and 2 if submitted before or on October 11.